

institutions that make us distinct and that provide those qualities that at times make less apparent the continuum between humans and other animals. One such quality, our considerable capacity for reciprocal altruism, will be essential as we tackle current challenges, including global climate change, in which individual and collective interests often appear to be in conflict.

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Interview with E. O. Wilson: Pioneering the field of sociobiology



CONCEPT CHECK 51.4

1. Explain why geographic variation in garter snake prey choice might indicate that the behavior evolved by natural selection.
2. Suppose an individual organism aids the survival and reproductive success of the offspring of its sibling. How might this behavior result in indirect selection for certain genes carried by that individual?
3. **WHAT IF?** Suppose you applied Hamilton's logic to a situation in which one individual is past reproductive age. Could there still be selection for an altruistic act?

For suggested answers, see Appendix A.

51 Chapter Review



➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkz9t.

CONCEPT 51.1

Discrete sensory inputs can stimulate both simple and complex behaviors (pp. 1140–1143)

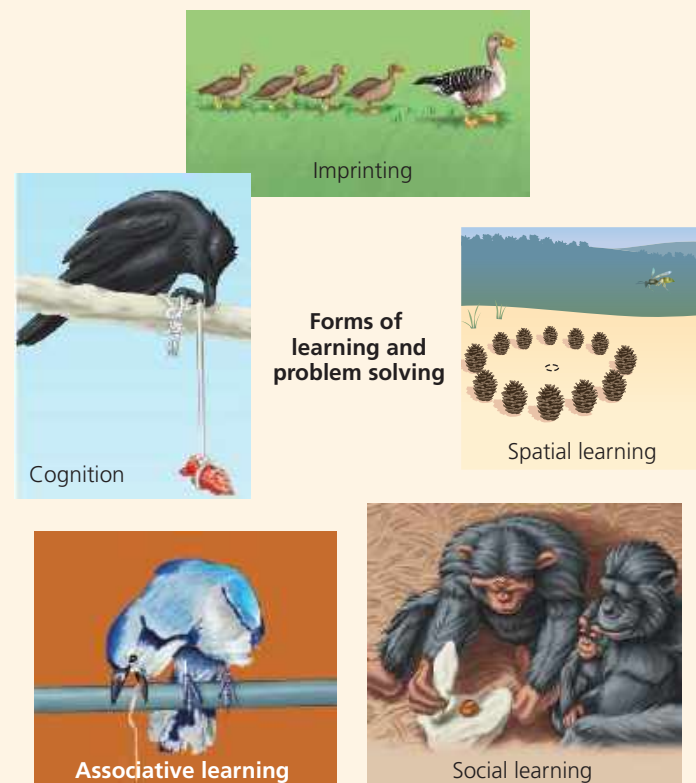
- **Behavior** is the sum of an animal's responses to external and internal stimuli. In behavior studies, proximate, or "how," questions focus on the stimuli that trigger a behavior and on genetic, physiological, and anatomical mechanisms underlying a behavioral act. Ultimate, or "why," questions address evolutionary significance.
- A **fixed action pattern** is a largely invariant behavior triggered by a simple cue known as a **sign stimulus**.
- **Migratory** movements involve navigation, which can be based on orientation relative to the sun, the stars, or Earth's magnetic field. Animal behavior is often synchronized to the circadian cycle of light and dark in the environment or to the seasons.
- The transmission and reception of **signals** constitute animal **communication**. Animals use visual, auditory, chemical, and tactile signals. Chemical substances called **pheromones** transmit species-specific information between members of a species in behaviors ranging from foraging to courtship.

? How is migration based on circannual rhythms poorly suited for adaptation to global climate change?

CONCEPT 51.2

Learning establishes specific links between experience and behavior (pp. 1143–1148)

- Cross-fostering studies can be used to measure the influence of social environment and experience on behavior.
- **Learning**, the modification of behavior as a result of experience, can take many forms, as summarized in the diagram that follows.



? How do imprinting in geese and song development in sparrows differ with regard to the resulting behavior?

CONCEPT 51.3

Selection for individual survival and reproductive success can explain diverse behaviors (pp. 1148–1154)

- Controlled experiments in the laboratory can give rise to interpretable evolutionary changes in behavior.
- An **optimal foraging model** is based on the idea that natural selection should favor **foraging** behavior that minimizes the costs of foraging and maximizes the benefits.
- Sexual dimorphism correlates with the types of mating relationship, which include **monogamous** and **polygamous** mating.